

Pelagic Species Trait Database: Morphometric Measurement Guide

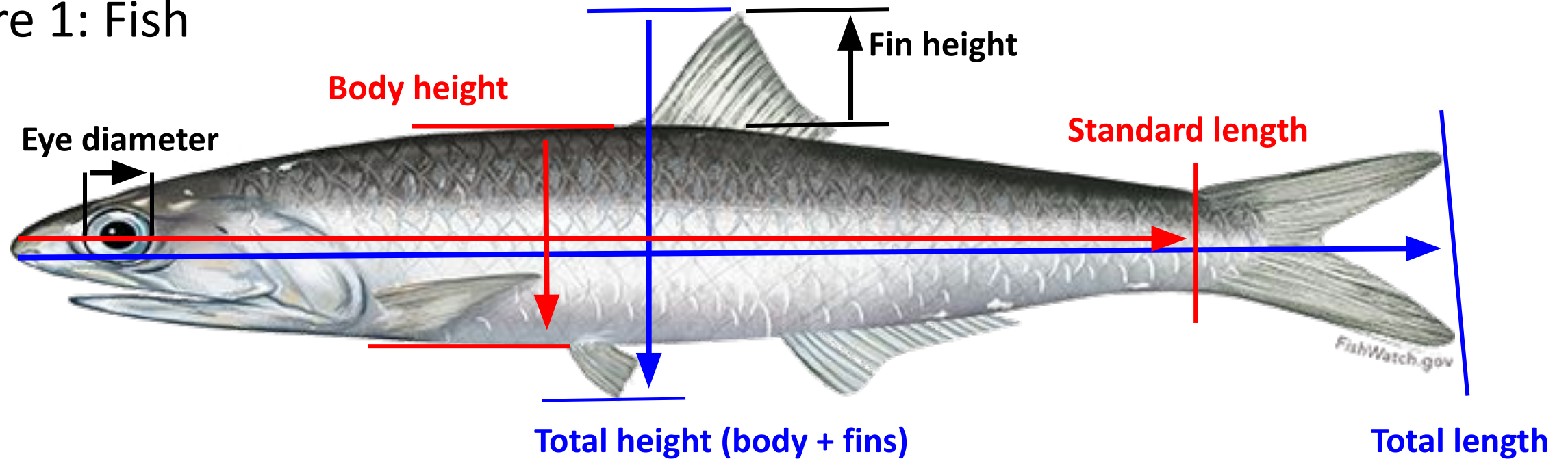
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Updated: 09/22/2023

Database measurement	Taxa	Measurement	Definition	Figure
Total length (TL)	fish	total length (TL)	Tip of the snout (or end of the longest jaw) to the end of the longest caudal lobe (Kahn et al 2004)	1,9
	cephalopod	total length	Tip of the tail (posterior end of mantle) to the end of longest arm; excludes the feeding tentacles which can retract (Roper and Voss 1983)	2,3,10
	crabs	total length	Tip of longest pereopod on one side of body to tip of pereopod on other side of body	6
	other crustaceans	total length	Rostral margin (e.g., base of the antennae/setae) to the posterior margin of the telson excluding setae (Morris et al 1988, Guerao et al 2006)	4,5
	gastropod	total length	Tip of the shell to the end of the body extended out of the shell	7
	other invertebrates	total length	Longest dimension (TL = SL)	8
Standard length (SL)	fish	standard length (SL)	Tip of the snout (or end of the longest jaw) along the lateral line to the base of the caudal fin (posterior limit of the hypural plate), where a groove forms (Kahn et al 2004, Howe 2002)	1
	cephalopod	mantle length (ML)	Tip of the tail (posterior end of mantle) to anterior most point of mantle (Roper and Voss 1983)	2,10
	octopus	mantle length (ML)	Posterior tip of the mantle to the midpoint between eyes ³¹	3
	crabs	carapace width (CW)	Distance between lateral spines of the carapace, at the widest point	6
	other crustaceans	standard length (SL)	Rostral margin to the base of the telson (posterior edge of 6th abdominal segment) (Morris et al 1988, Isaacs et al 1969)	4,5
	gastropod	shell length (SL)	Lateral distance from anterior to posterior on the shell (e.g., longest dimension) (Obaza and Ruehl 2013)	7
	other invertebrates	standard length	Longest dimension (TL = SL)	8
Eye diameter (eye)	all	eye diameter	Diameter of the eye, excluding tissue that contains the eye	1-6,10

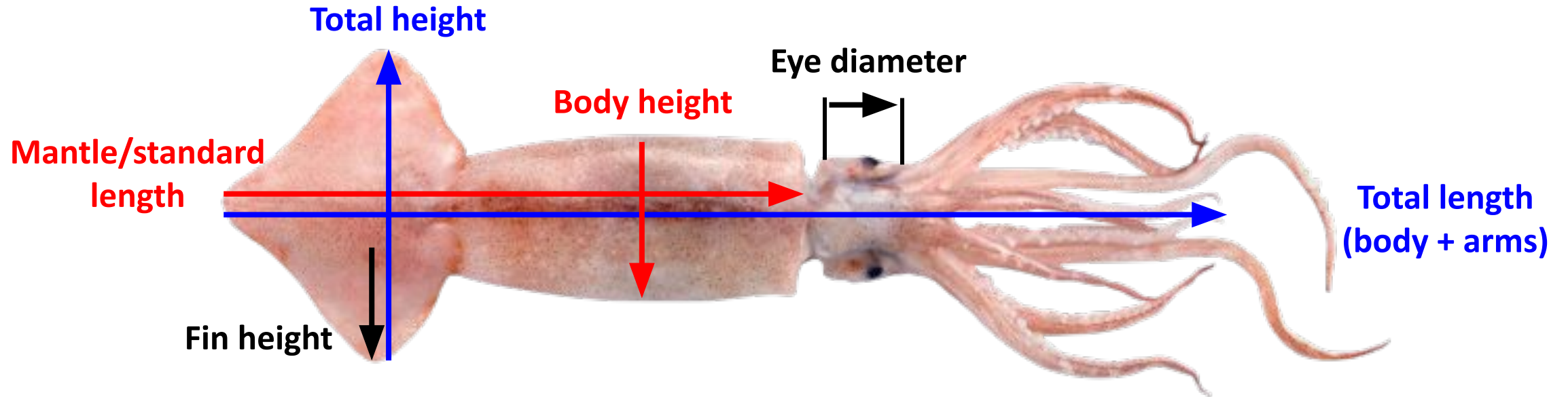
Database measurement	Taxa	Measurement	Definition	Figure
Total height (TH)	fish	total height (TH)	Tip of dorsal fin to tip of pelvic fin (or widest dimension to tips of fins), with fins fully extended (body height + fin lengths)	1,9
	cephalopod	total width	With fins (e.g., squid): diameter of mantle and fins combined, at the widest point; tip to tip of stabilizing fins. No fins (e.g., octopus): TH = BH	2,3
	crabs	total width	In dorsal view, anterior side of the folded claws to the posterior end of the carapace (carapace length plus folded claws)	6
	other crustaceans	total height	In lateral view, distance from dorsal-most margin to ventral-most margin at the widest point, including limbs	4,5
	gastropod	shell width (SW)	Shell width at widest point, including body out of shell (shell + appendages)	7
	other invertebrates	total height	Second longest dimension (TH = BH)	8
Body height (BH)	fish	body height (BH)	Base of the dorsal fin to base of pelvic fin; deepest part of the fish	1,9
	cephalopod	mantle width (MW)	Diameter of the mantle at the widest point (not including fins)	2,3
	crabs	carapace length (CL)	In dorsal view, distance from anterior to posterior ends of the carapace	6
	other crustaceans	carapace width (CW)	In lateral view, distance from dorsal side of the carapace to the base of the limbs	4,5
	gastropod	shell width (SW)	Shell width at widest point	7
	other invertebrates	body height	Second longest dimension (TH = BH)	8
Fin height (FH)	fish	fin height	Dorsal fin, from base to tip	1,9
	cephalopod	fin height	Stabilising fin, from base to tip on one side	2

Figure 1: Fish



- **Total length (TL)**: Tip of the snout (or end of the longest jaw) to the end of the longest caudal lobe.
 - If caudal fin is present, but not visible: SL = NA.
- **Standard length (SL)**: Tip of the snout (or end of the longest jaw) along the lateral line to the base of the caudal fin (posterior limit of the hypural plate), where a groove forms.
- **Total height (TH)**: Tip of dorsal fin to tip of pelvic fin (or widest dimension to tips of fins), with fins fully extended (body height + fin lengths).
 - If pelvic fin or dorsal fin is present, but not visible, TH = NA.
- **Body height (BH)**: Base of the dorsal fin to base of pelvic fin; deepest part of the fish
- **Fin height (FH)**: dorsal fin, from base to tip

Figure 2: Squids



- **Total length (TL):** Tip of the tail (posterior end of mantle) to the end of longest arm; excludes the feeding tentacles which can retract.
- **Standard length (SL):** Tip of the tail (posterior end of mantle) to anterior most point of mantle.
- **Total height (TH):** Diameter of mantle and fins combined, at the widest point; tip to tip of stabilizing fins.
 - If fins are present, but not visible, TH = NA.
- **Body height (BH):** Diameter of the mantle at the widest point (not including fins)
- **Fin height (FH):** Stabilizing fin, from base to tip on one side.

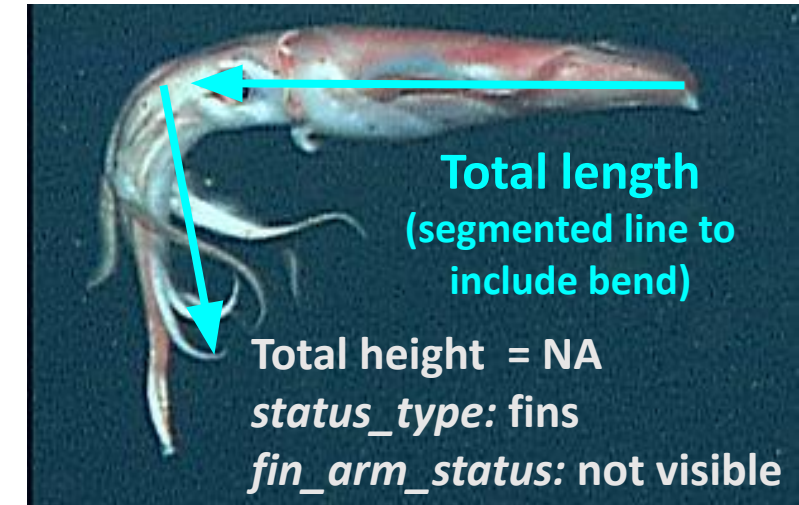
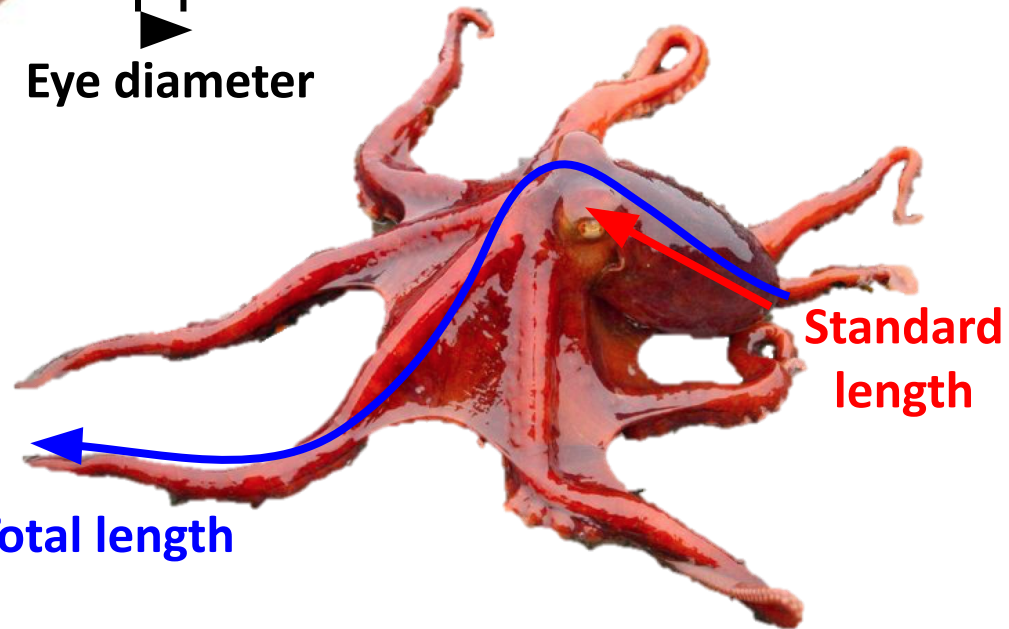
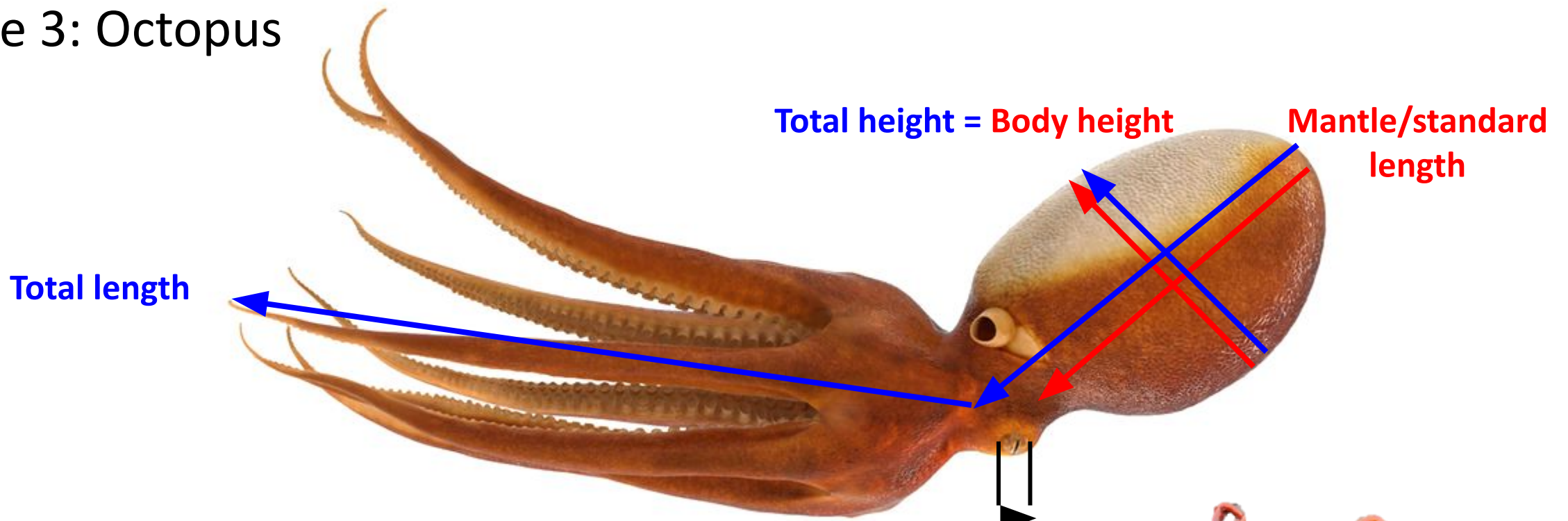
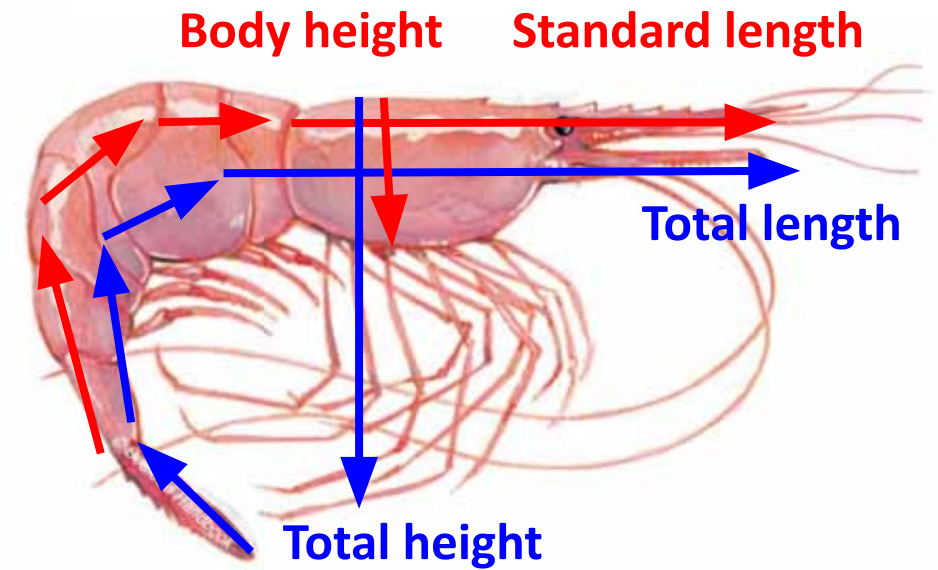
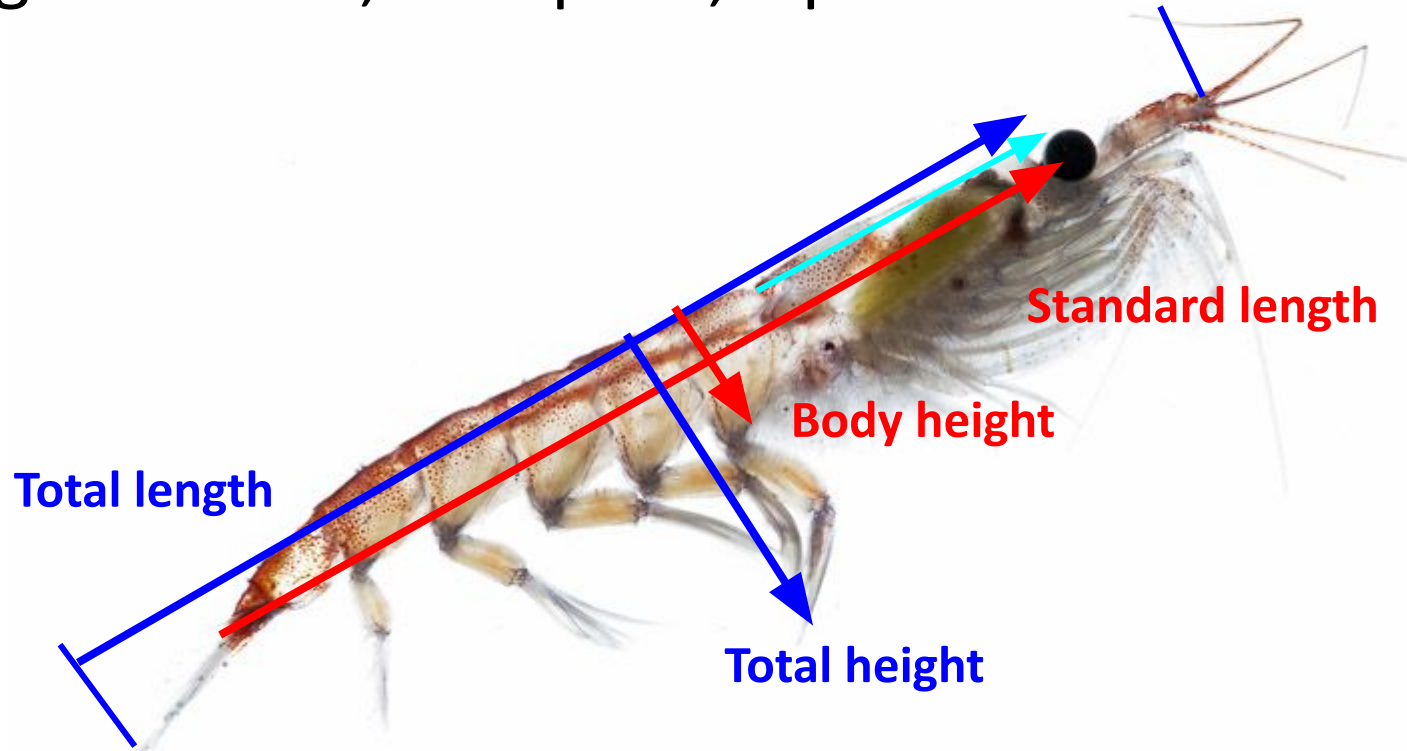


Figure 3: Octopus



- **Total length (TL):** Tip of the tail (posterior end of mantle) to the end of longest arm; excludes the feeding tentacles which can retract.
- **Standard length (SL):** Posterior tip of the mantle to the midpoint between eyes.
- **Total height (TH):** No fins, TH = BH
- **Body height (BH):** Diameter of the mantle at the widest point
- **Fin height (FH):** No fins, FH = 0

Figure 4: Krill, decapods, squat lobsters



- **Total length (TL):** Rostral margin (e.g. base of the antennae/setae) to the posterior margin of the telson excluding setae
- **Standard length (SL):** Rostral margin to the base of the telson (posterior edge of 6th abdominal segment) Note: example on right has a longer rostrum
- **Total height (TH):** In lateral view, distance from dorsal-most margin to ventral-most margin at the widest point, including limbs
- **Body height (BH):** In lateral view, distance from dorsal side of the carapace to the base of the limbs
- **Fin height (FH):** No fins, FH = 0

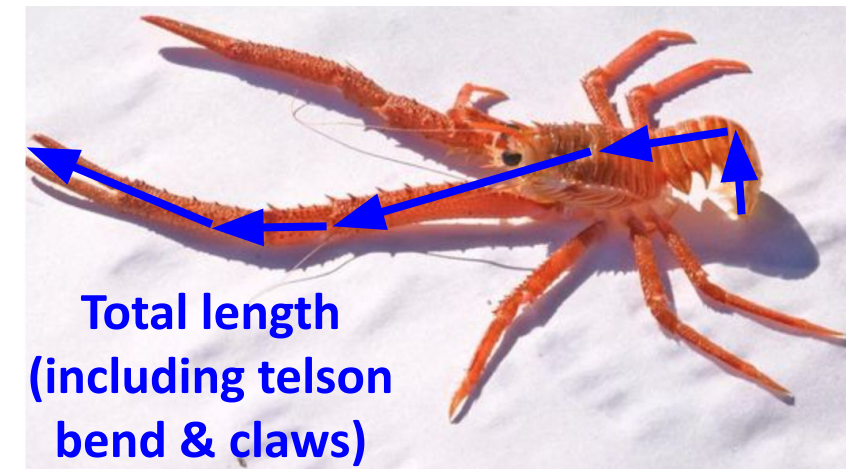
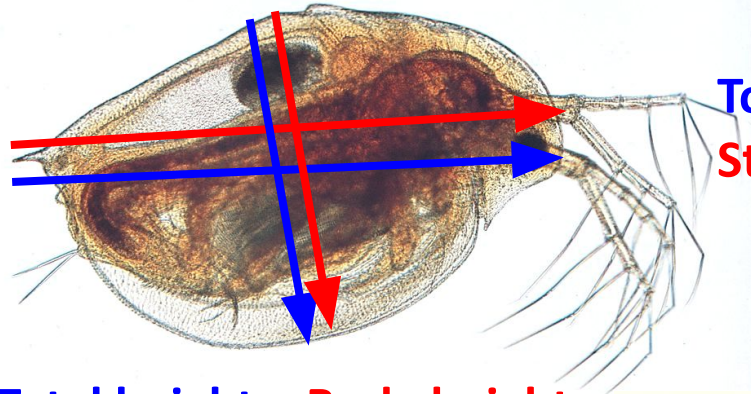


Figure 5: Other crustaceans

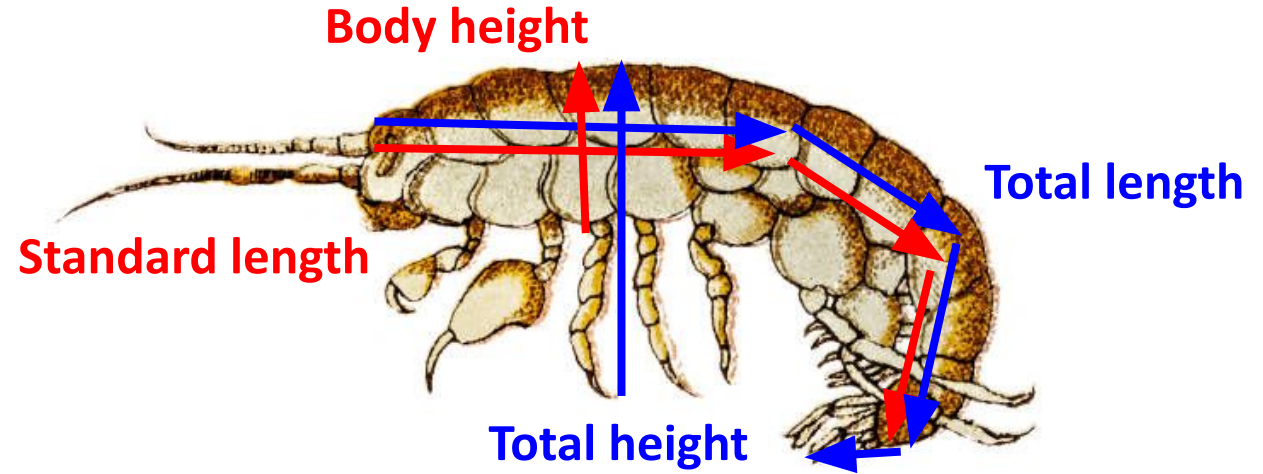
Cladocerans



Total length =
Standard length

Total height = Body height

Amphipods



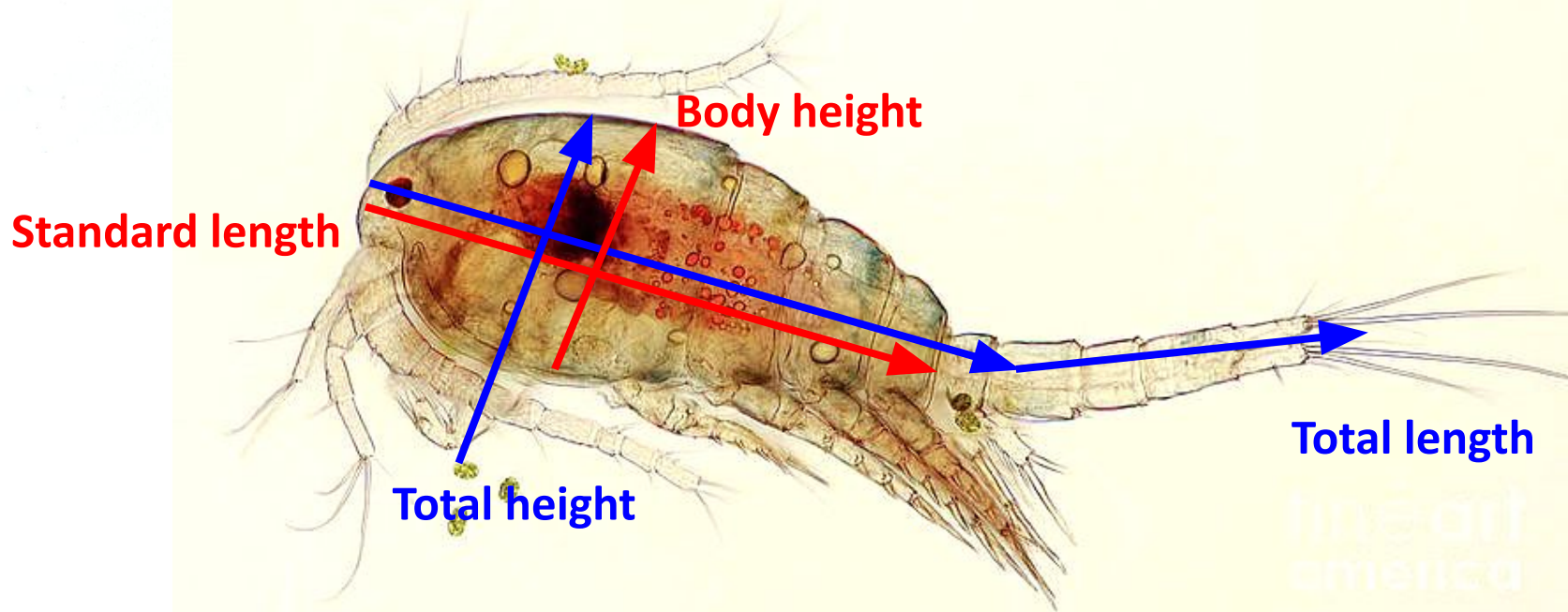
Body height

Standard length

Total height

Total length

Copepods



Body height

Standard length

Total height

Total length

line art
america

Figure 6: Crabs

- **Total length (TL):** Tip of longest pereopod on one side of body to tip of pereopod on other side of body.
- **Standard length (SL):** Distance between lateral spines of the carapace, at the widest point.
- **Total height (TH):** In dorsal view, anterior side of the folded claws to the posterior end of the carapace (carapace length plus folded claws).
- **Body height (BH):** In dorsal view, distance from anterior to posterior ends of the carapace.
- **Fin height (FH):** No fins, $FH = 0$

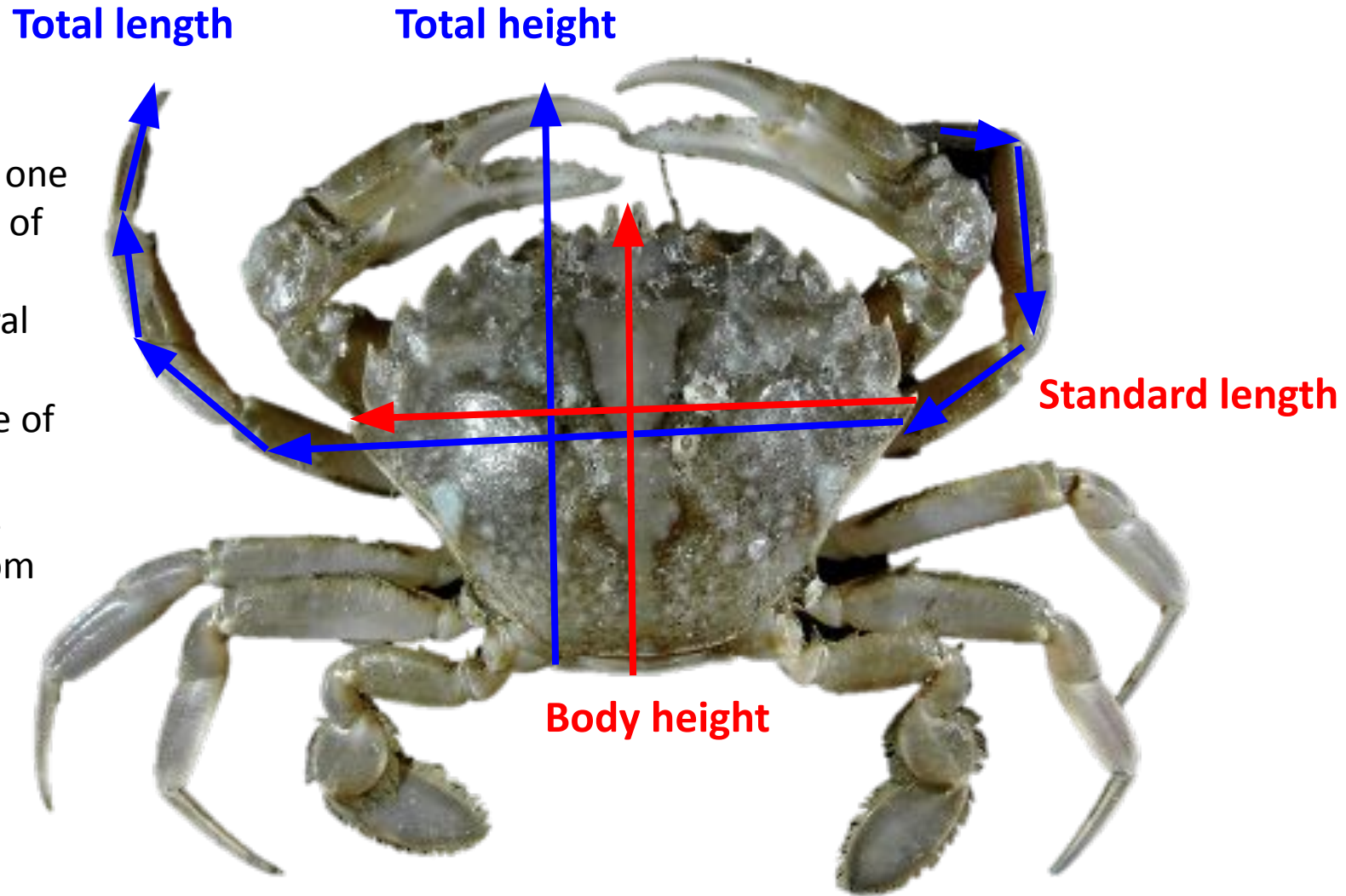
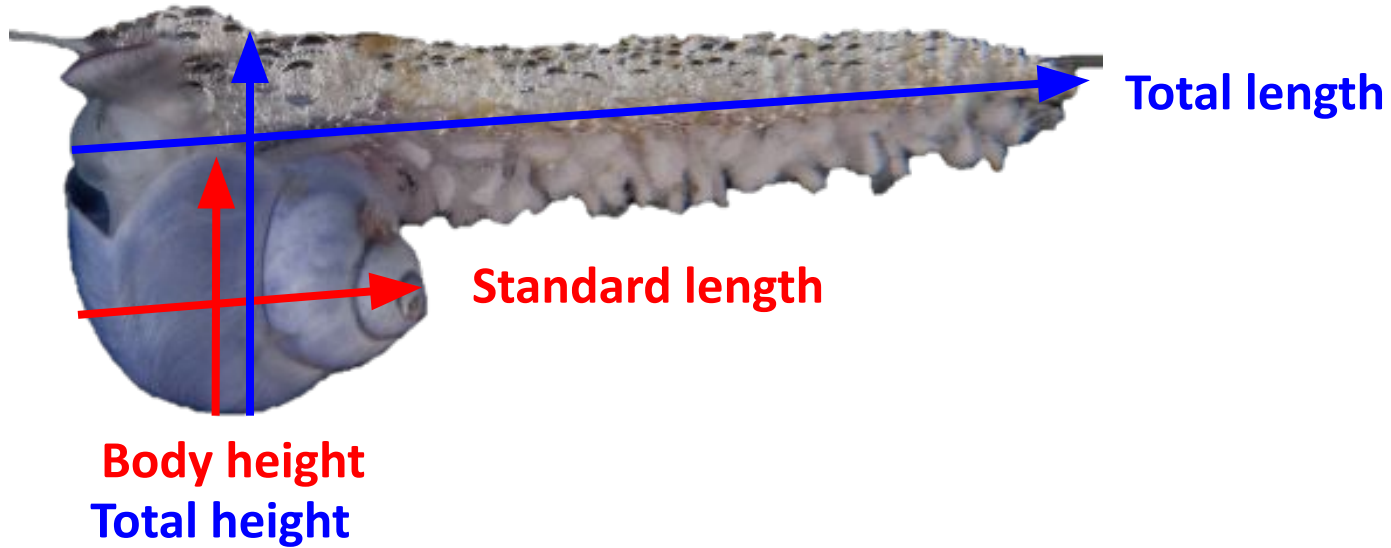
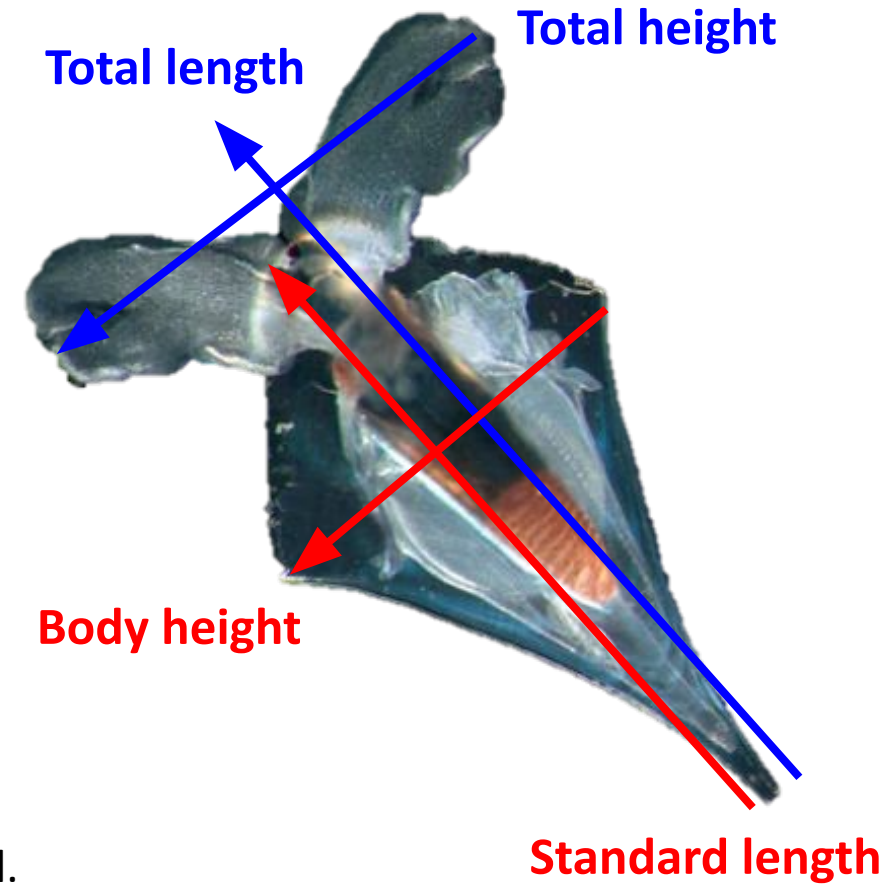


Figure 7: Gastropod



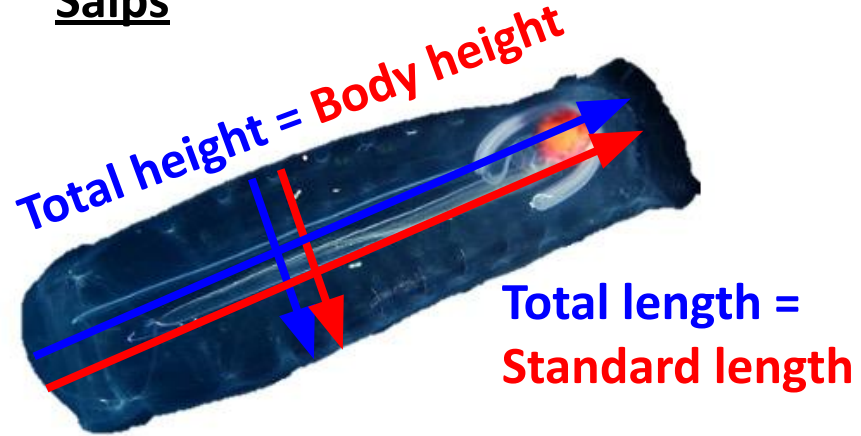
Pteropods



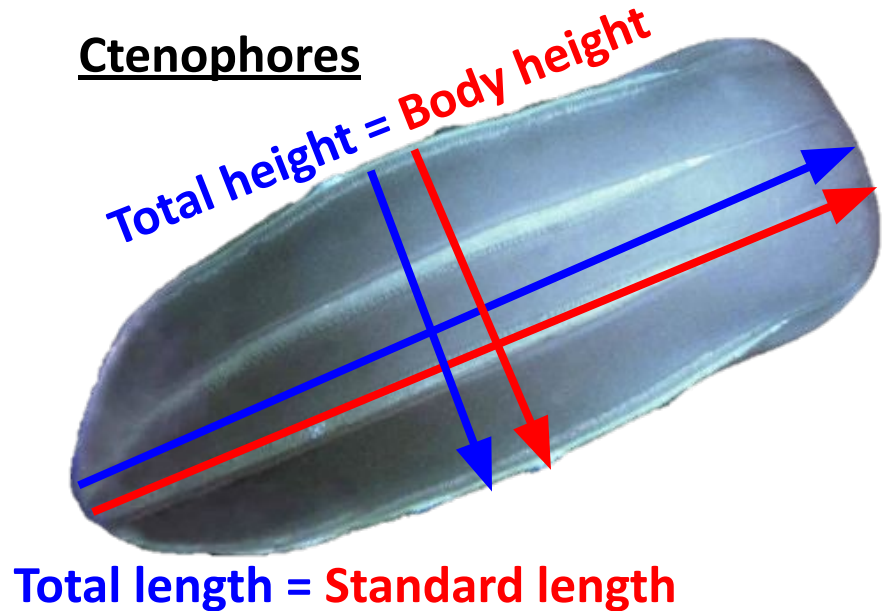
- **Total length (TL):** Tip of the shell to the end of the body extended out of the shell.
- **Standard length (SL):** Lateral distance from anterior to posterior on the shell (e.g., longest dimension).
- **Total height (TH):** Shell width at widest point, including body out of shell (shell + appendages).
- **Body height (BH):** Shell width at widest point.
- **Fin height (FH):** No fins, FH = 0.

Figure 8: Gelatinous invertebrates

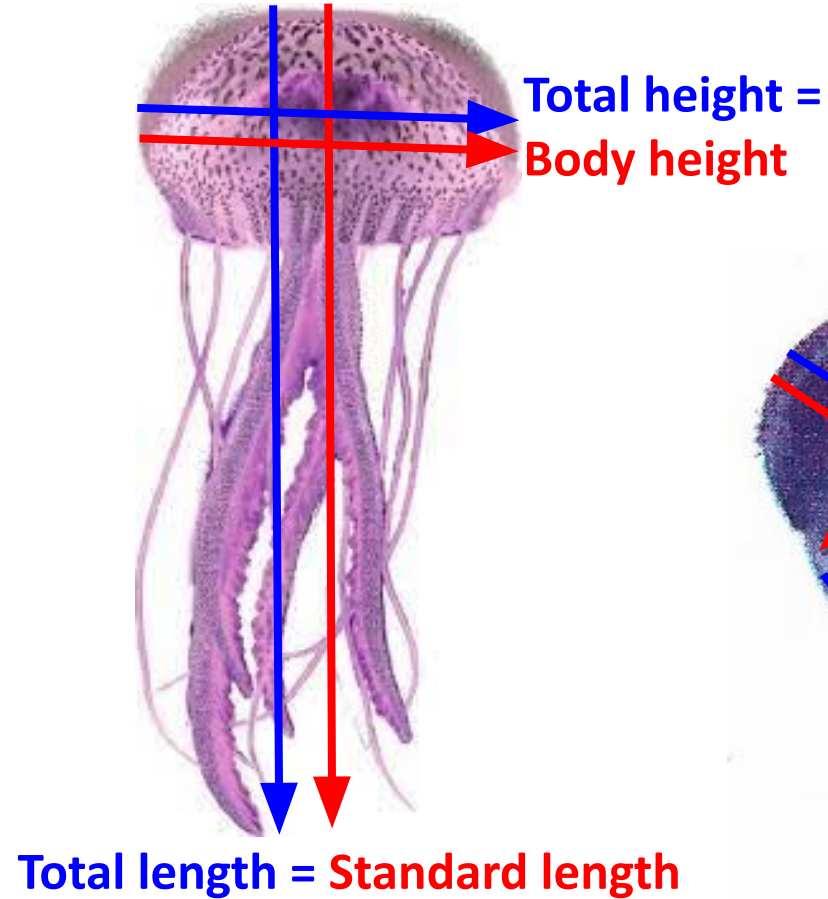
Salps



Ctenophores



Scyphozoans or hydrozoans



Cydippid ctenophores

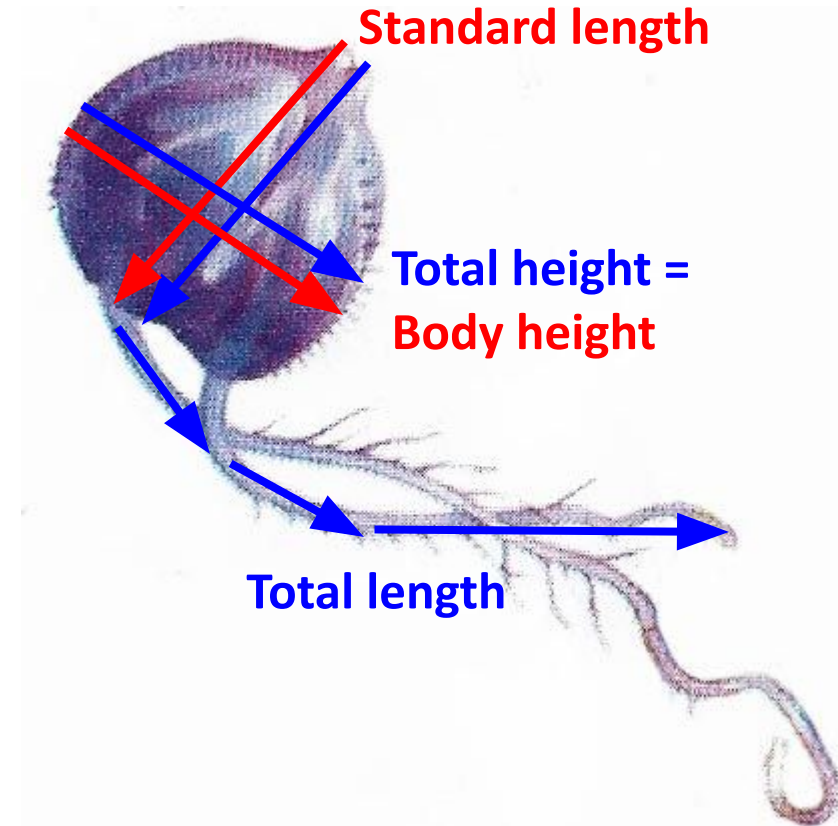


Figure 9: Skates & rays (special case)

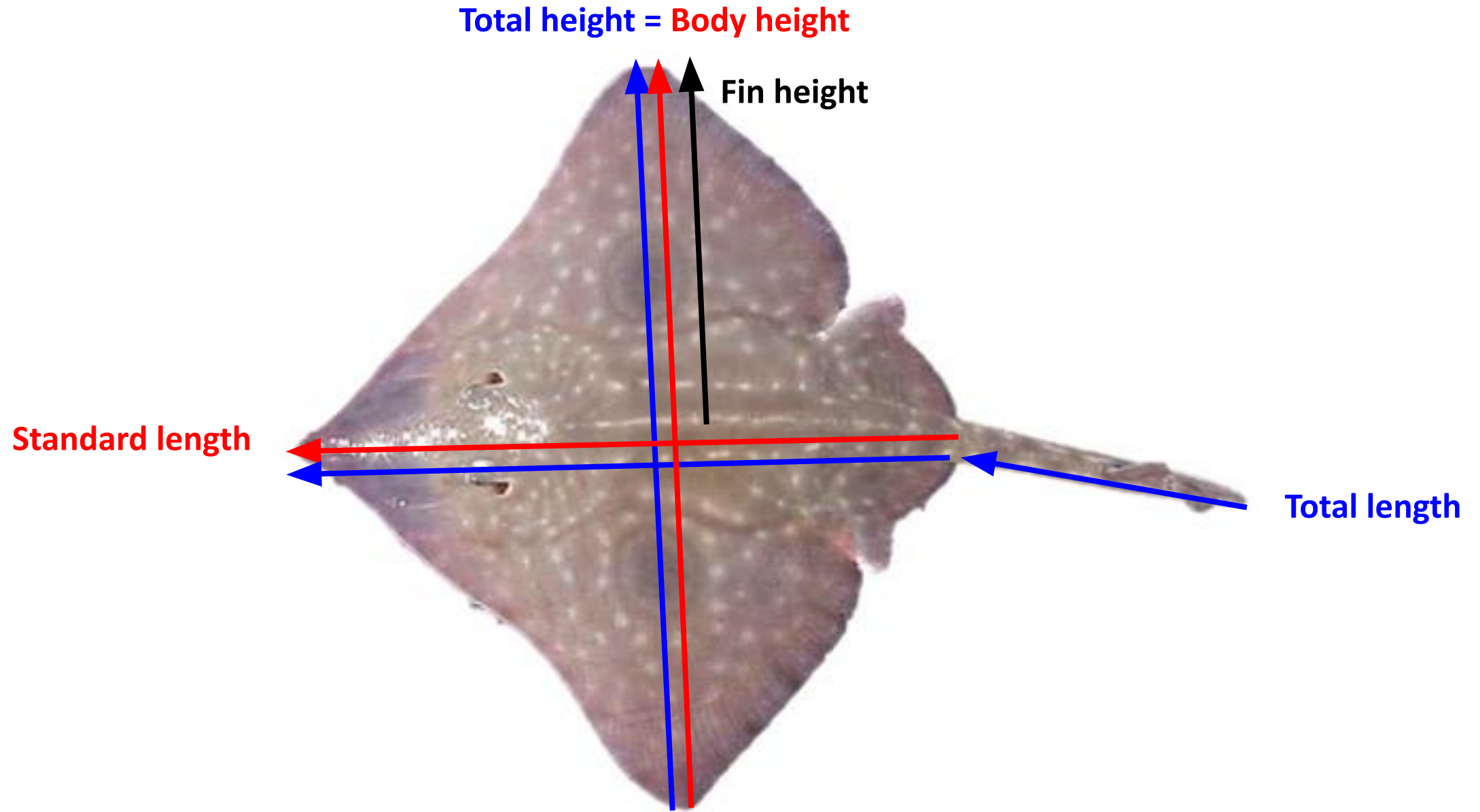
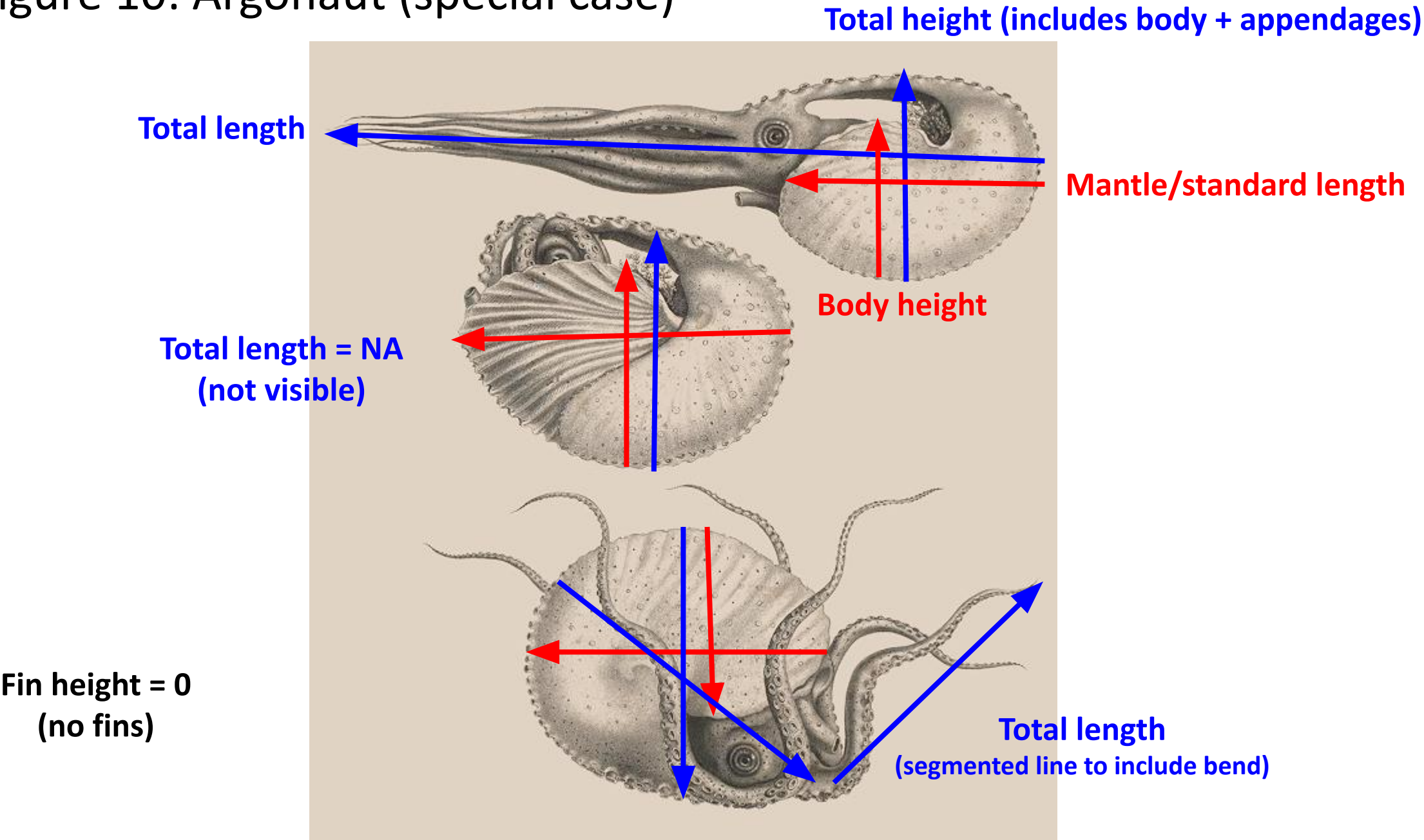


Figure 10: Argonaut (special case)



Measurement definition references:

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